

Censorship as Bitcoin Grows

A Powerful Dummy's guide to censoring Bitcoin

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Intro

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- the public ledger is a public ledger

Censorship and Bitcoin

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- the public ledger is a public ledger
- happens on other chains

- Feather Forking (is it enough to control a non trivial minority of hashrate?)

Outline

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Classical Approach: Feather Forking

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- $\implies$ 4% risk to miner for including a censored transaction

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Feather Forking

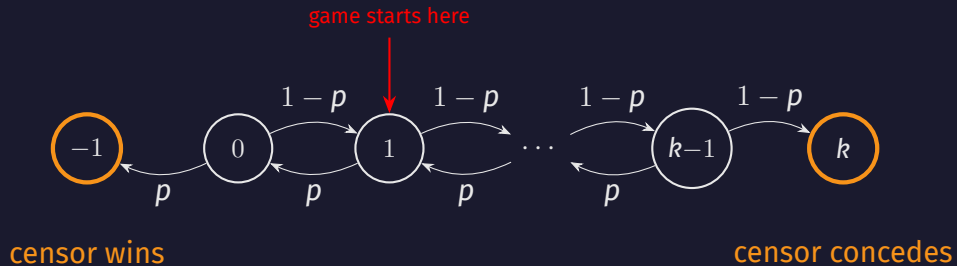
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- I roughly agree
- I use different models - fix hashrate of each agent, run the game and they decide their strategy

Random Walk Model



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- This is a finite difference equation (like discrete 2nd order ODE)
- Add a boundary condition $W(-1) = 1$ and $W(k) = 0$ and this becomes solvable

$$W(x) = \frac{\left(\frac{p}{1-p}\right)^k - \left(\frac{p}{1-p}\right)^x}{\left(\frac{p}{1-p}\right)^k - \left(\frac{p}{1-p}\right)^{-1}} \quad (1)$$

Probability of a Successful Reorg

p	$W(1), k = 2$	$W(1), k = 5$
0.10	0.0110	0.0123
0.20	0.0476	0.0623
0.30	0.1139	0.1786
0.45	0.2691	0.5278
0.60	0.4737	0.8797

Expected Loss for Compliant Mining

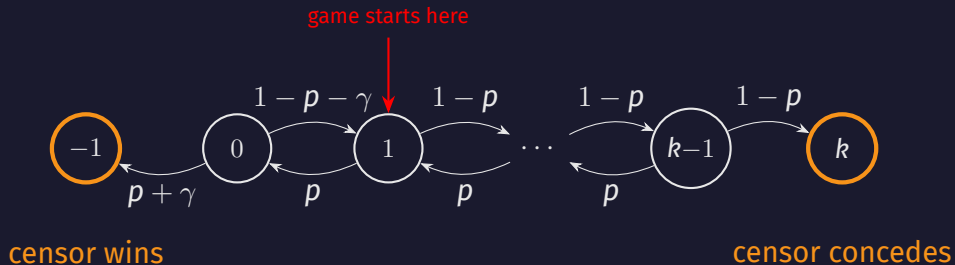
Blocks missed by the compliant pool versus mining the chain tip all along:

p	$C(1, 0), k = 2$	$C(1, 0), k = 5$
0.1	0.0978	0.4630
0.2	0.1814	1.0442
0.3	0.2355	1.6170
0.45	0.2404	1.5170
0.6	0.1662	0.3473

Minimum fee required to make mining the noncompliant block rational:

p	$f^*, k = 2$	$f^*, k = 5$
0.10	0.0112	0.0128
0.20	0.0512	0.0717
0.30	0.1335	0.2561
0.45	0.3902	1.5711
0.60	0.9711	10.9032

Random Walk Model 2 some miners rationally join censor



- Costly to miners - may require subsidies

Feather Forking Analysis

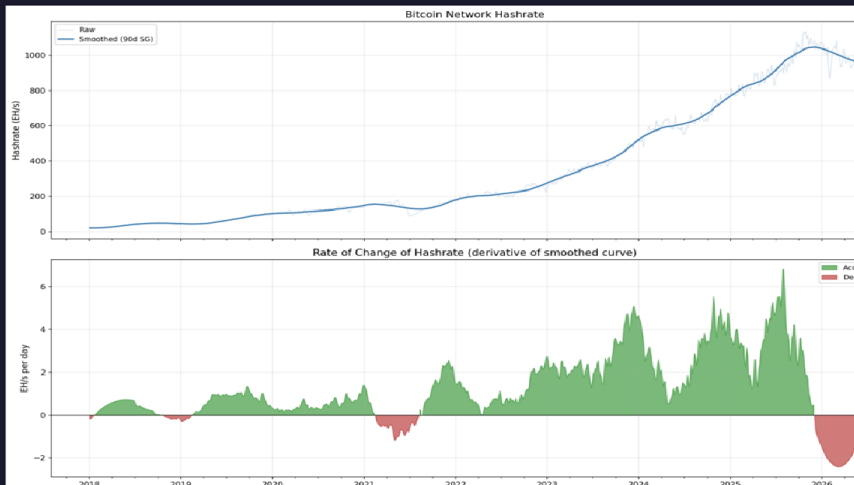
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- Drawn out races could be counterproductive

Fortifying Feather Forks

Hashrate stockpile : Napkin Math



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- Energy draw: 8–12 GW $\approx 2\%$ of US grid — not trivial, must be geographically distributed
- At $\$0.10/\text{kWh}$: $\$1\text{M/hr}$, $\$24\text{M/day}$ — expensive sustained, **cheap if used only for hours**

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- As subsidy decreases and price doesn't double, could become more feasible

Hash rental and pool operating

- economically motivated miners are economically motivated

Running a pool

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- Calculus entirely depends on how many miners can be wooed by higher payouts

- **Full Pay Per Share** - pool pays miner for partial proofs of work

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- FPPS pools are attractive during race, miners will seek out highest paying pool and point there

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- Misaligned incentives during race
- Very hard to detect conclusively

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- This still does not prevent some block withholding

Big Hard Problems With Soft Answers

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- No more happy path for stakeholders in general

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- What possible responses are on the table?

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- In response to disruption?

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- options exist – INVALIDATEBLOCK, checkpointing, temporary sentinels, "bunker mode"

- Significant Economic nodes on incompatible chains $\implies$ existential crisis

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- Economic nodes committing to a side $\implies$ game of chicken
- Domiciled institutions are committed to your side (no choice) $\implies$ the rest of the network has no choice
- (Willingness to blow everything up rather than accept censorial overlords is a real and available very Thomas Schelling-coded option, if economic power is available.)

- Ultimately, miners need economic nodes to enforce will

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- If you have economic control... . why not just wield that directly?

Tainting: How to leverage institutional adoption

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- Softer touch than reorging

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- Institutions must certify they didn't give anything of value when receiving the tainted coins

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- lesser coin with less demand

The Taint Paradigm: Reframing Fungibility at Scale

At the end of the day, institutional demand is the load-bearing element, quietly doing all the work here – and that changes everything

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- Compliance burden – Yes - not deep, but wide

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- Use DeFi (Thorchain, for example) to anchor market prices early

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- Stage 4: Inch retroactive, apply market pressure to exchanges to do due diligence

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- Clear conscience for privacy enthusiasts - coinjoin without helping North Korea launder money
- Permanent: bad actors can't just RBF out of it
- End game is a button to freeze (at our discretion) any UTXO we don't like - very useful tool!

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- Maybe other people not happy
- Unclear what happens if other jurisdictions try it
- May cause parties who don't like this power wielded to avoid Bitcoin entirely (works too well! giving US too much arbitrary power)

Thank you!